Uğur Tepe

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Education

Master of Science (M.Sc.), Physics 2023 – Present
Heinrich-Heine-University *Düsseldorf*
Focus on Majorana fermions and topological quantum computation

Bachelor of Science (B.Sc.), Physics 2018 – 2023
Heinrich-Heine-University *Düsseldorf*
Thesis: Adaptively measuring quantum expectation values using the empirical Bernstein stopping rule
Quantum Technology Research Group , Prof. Dr. Martin Kliesch

Experience

Working Student in Finance-, Data- and AI-Services Spring 2024 – Present
ARAG Versicherung *Düsseldorf*

- Maintaining and assisting in development of new software
- Managing and testing new software
- Market research and analysis

Working Student in Event Coordination Spring 2020 – Spring 2022
Klavier-Festival Ruhr *Essen*

- Logistics of talent and gear
- Coordination and planning of events

Conferences/ Projects

DPG Spring Meetings 2024 Berlin
Poster: Presentation on Empirical Bernstein Stopping in Quantum Algorithms *2024*

Publications

1. Gresch, Alexander, Uğur Tepe, and Martin Kliesch. "Reducing the sampling complexity of energy estimation in quantum many-body systems using empirical variance information." Journal of chemical theory and computation (2025)

Languages

Languages: German (Native), Turkish (Native), English (Fluent)

Technical Skills

Programming Languages: Python, LaTeX, C/C++, Java, HTML/CSS
Libraries: NumPy, Pandas, Matplotlib, Scipy